Fiscal Year:	FY 2024	**Task Last Updated:**	FY 07/08/2024
PI Name:	Wyatt, Sarah E Ph.D.		
Project Title:	Spaceflight Alters Post-Transcriptional Regulation		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	45701	**Congressional District:**	15
Comments:			
Project Type:		**Solicitation / Funding Source:**	
Start Date:	09/01/2019	**End Date:**	08/31/2024
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA KSC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 08/31/2024 per NSSC information (Ed., 7/27/24). NOTE: End date changed to 05/31/2024 per NSSC information (Ed., 5/2/23). NOTE: End date changed to 05/31/2023 per NSSC information (Ed., 6/6/22).		
Key Personnel Changes/Previous PI:	The postdoctoral fellow accepted a new position at Kennedy Space Center and left OU on June 2022.		
COI Name (Institution):	Perera, Imara Ph.D. (North Carolina State University)		
Grant/Contract No.:	80NSSC19K1481		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Plants will be a crucial component for astronaut health and well-being during any long-distance spaceflight or colonization mission: as a source of food, for replenishing water and purifying air as well as physiological and psychological comfort. The challenge is to understand how plants respond to the spaceflight environment to enable plants to thrive in potentially hostile environments. As technologies have advanced for global gene expression, transcriptome level research has become an integral part of experiments and provides knowledge of gene expression resulting from the conditions of spaceflight. Steady state transcript abundance quantified by RNASeq is frequently used as measure of gene expression with the implicit assumption that transcriptional changes upon a treatment (such as a stress) are indicative of the downstream response. However, changes in mRNA abundance do not necessarily lead to corresponding changes in protein, and transcript and protein abundance are not highly correlated. The data from spaceflight experiments Biological Research in Canisters-20 (BRIC20) (Principal Investigator: Wyatt) and Plant Signaling (Principal Investigator: Perera) suggest that post-transcriptional regulation plays an integral role in gene expression differences between spaceflight and ground controls. The differential expression, abundance, and phosphorylation of translational machinery in spaceflight leads to the obvious questions: What genes are being post transcriptionally regulated and by what mechanism(s)? To answer these questions, principal investigators Wyatt and Perera plan to combine expertise and resources to generate compatible multi-omics datasets and provide a comprehensive picture of transcriptional and post transcriptional regulation. This integrated approach will help answer fundamental questions regarding plant adaptation to spaceflight and this proposal is aligned with sub-topic PL-A of Appendix B ("Study plant growth in spaceflight environments and physiological responses to the multiple stimuli encountered in those environments"). The novel datasets and analyses will address questions outlined in Research Emphasis 2 of the Space Biology Science Plan 2016-2025 (Plant Biology Studies in Support of Space Exploration), specifically "(a) Answer basic questions about how plants respond to changes in gravity and other environmental factors associated with spaceflight and (b) Build on past plant research to provide a better understanding of physiological responses of plants to spaceflight, providing new knowledge that can facilitate the development of a bioregenerative life support system." Analyses will also address Research Emphasis 4: molecular and cellular biology, specifically "(a) Studies designed to determine how spaceflight alters gene expression at the transcriptomic, metabolomics and proteomics levels in the different tissues or cell types within the same organism, and how these changes impact the organism's overall health during space travel."
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Plants are critical for human life and well-being on Earth. However, many areas of the world have limited-resources for adequate plant growth to support life. For plants to thrive in these less than hospitable environments, the challenge is to understand how plants respond to a variety of stresses and be able to mitigate those responses. Researchers often use the transcriptome, those genes transcribed from DNA, as a proxy for protein produced because of the speed and relative ease of acquisition and analysis. Full analysis of proteins requires much more tissue and time but also only provides a subsection of the protein complement due to current limitation of techniques. Here, we use analysis of polysome associated RNAs, those transcripts that are associated with polysomes, the translational machinery of the cell. The research proposed will also generate datasets for the full transcriptome and small, regulatory RNAs. The goal is to use the three datasets to provide a more comprehensive picture of gene expression and its regulation as affected by spaceflight. The project will answer basic questions about how plants respond to changes in gravity and other environmental factors associated with spaceflight, and build on past plant research to provide a better understanding of physiological responses of plants to spaceflight, providing new knowledge that can facilitate the development of a bioregenerative life support system. In addition, it will also add to our understanding of how spaceflight alters gene expression at the gene transcript and protein levels in both roots and shoots. Overall, it will lead to findings that may provide potential targets for environmental or genetics adaptions for plant growth for food and other uses during spaceflight impacting astronaut help and well-being. In addition, it will provide a deeper understand of how plants respond to environmental factors that may also benefit plant growth on Earth on marginal lands or in limited spaces to increase production.
Task Progress:	Summary of work in Year 4 – Data analyses Year 1 of the project saw the completion of several of the preflight objectives on multiple avenues of inquiry. Year 2 focused on preflight verification experiments Science (S) and Experiment (E) Verification Tests (VT) and launch readiness reviews and final launch preparation, culminating in launch aboard SpaceX-22 on June 3, 2021. Year 3 saw the partial return of the flight samples, leading to a discussion with Kennedy Space Center about options and ultimately resulting in a second flight of the experiment with all samples returned successfully. The second flight of the experiment launched on SPACE-X 24 on Dec 21, 2021. After return, all samples were prepared for sequencing. Data Return and Analyses (Year 4). The total mRNA, polysomal mRNA, and small RNA were submitted to the Genomics Sciences Laboratory at NCSU in the summer of 2022. The sequencing facility initially estimated that sequence data would be made available by September 2022. Unfortunately, sequences were not fully returned until January 23rd of 2023. In spite of these setbacks, considerable analyses have been completed. EVT analyses. Three root samples were prepared from the experimental validation test (EVT) to confirm the viability of the proposed extraction methods. These data included polysomal and total mRNA from root tissue grown on the veggie hardware on earth. Both the total mRNA and polysomal extractions produced libraries of aligned reads of equivalent sizes (welch two-sample t-test, $p = 0.927$). Of the 25,102 unique loci returned after alignment, 93% of genes detected had transcripts present in both fractions. The relative proportion of transcripts followed a normal distribution. This distribution was not appreciably different when comparing reads of various lengths indicating that neither method was severely biased towards longer or shorter sequences. These trends in expression are also conserved across nuclear chromosomes. The few areas with higher polysomal representation are randomly spread across chromosomes with no apparent bias towards the chromosomal region. Taken together these results show that both the total mRNA and polysomal mRNA fractions produce high-quality read libraries with the equivalent ability to detect transcripts regardless of transcript length or the location of the encoding loci. Taken together these results show that both the total mRNA extraction and polysomal purification were capable of producing high-quality read libraries with no apparent bias towards any read type or read chromosomal location. The results from the EVT let us confidently analyze the flight data without the need for correction of the raw data. Flight Analyses. The delay in sequencing has set back the analyses of the data but some results can already be seen. The goals of this project are to identify the post-transcriptional regulation of RNA in space. These differences would not be visible when only looking at ground transcriptomes as compared to flight transcriptomes but would have a substantial impact on the plant's response to spaceflight. To date, all sequence data from the sRNA, polysomal RNA, and total mRNA have been trimmed, aligned, and quantified. Trimming uses TrimGalore under base parameters for quantity

	trimming was used to remove primers and low-quality reads. STAR and the Araport 11 genome annotation were used for read alignment. The sRNA is being normalized separately due to substantial differences in the library features between small RNA and mRNA extractions. Because of the consistency in libraries seen in the EVT, total and polysomal mRNA were normalized together for each tissue. Roots and stems were normalized separately, again due to major divergence in RNA populations. Before performing the differential expression, general features of the libraries normalized together were calculated and compared. This was done to preliminarily check for outlier groups as low quantities of successfully aligned reads could cause divergence in mean expression calculations in the downstream analysis. Minor differences in library size are corrected for during the analysis but large outliers (libraries of exponentially different sizes) would need to be handled with more severe corrections for differences with potential outlier removal. There were no major outliers in the relative library sizes for either stems or roots and the analysis could continue without any deviations from the plan. Even from the early analyses, we can see that the data produced will be able to address many of these key questions. The principle coordinate plot (PCoA) shows how samples are related to each other. Samples with a high degree of similarity are closer to each other while samples with high divergence will be farther away from each other on the graph. The PCoA from the flights shows that both flights have similar populations of total mRNA but highly divergent polysomal m mRNA in roots. The internal grouping of each set of flight replicates indicates that differences between flights were not due to outliers pushing the average but true biological signal. This finding is also confirmed by the initial breakdown of the differential expression results. Between the flight and ground samples, 37% of the DE genes were significant and regulated in the same direction in the total mRNA fraction from the first and second flights. The polysomal mRNA showed only 27% consistency in gene regulation between flights. These results further support our thesis that regulatory mechanisms controlling translation are not accessible when looking at only the transcriptome. The reason for these deviations will be more understandable following the contrast analysis between the mRNA and polysomal RNA. The contrast will show both how genes are being differentially regulated between mRNA and psRNA as well as the degree of similarity between the flights. The ability of the sRNA to regulate these observed shifts will then be quantified giving a deeper understanding of the mechanisms behind RNA regulation in space.
Bibliography Type:	Description: (Last Updated: 11/25/2025)
Abstracts for Journals and Proceedings	Canaday ELJ, Land ES, Meyers AD, Perera IY, Wyatt SE. "From transcription to translation, new levels of plant regulation visible through parallel sequencing of polysomal, messenger and microRNA from the APEX-07 spaceflight." 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. Abstracts. 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Dec-2024
Abstracts for Journals and Proceedings	Canaday ELJ, Land ES, Meyers AD, Perera IY, Wyatt SE. "New depths of plant regulation in spaceflight through profiling of miRNA, mRNA, and ribosome-associated mRNAs." ELGRA 2024, European Low Gravity Research Association Biennial Symposium and General Assembly, Liverpool UK, September 3-6, 2024. Abstracts. ELGRA 2024, European Low Gravity Research Association Biennial Symposium and General Assembly, Liverpool UK, September 3-6, 2024. , Sep-2024
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Articles in Peer-reviewed Journals	Barcenilla BB, Meyers AD, Castillo-González C, Young P, Min JH, Song J, Phadke C, Land E, Canaday E, Perera IY, Bailey SM, Aquilano R, Wyatt SE, Shippen DE. "Arabidopsis telomerase takes off by uncoupling enzyme activity from telomere length maintenance in space." Nat Commun. 2023 Nov 29;14:7854. https://doi.org/10.1038/s41467-023-41510-4 ; PMID: 38030615; PMCID: PMC10686995 , Nov-2023